

Traffic Flow

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December 31st 2022

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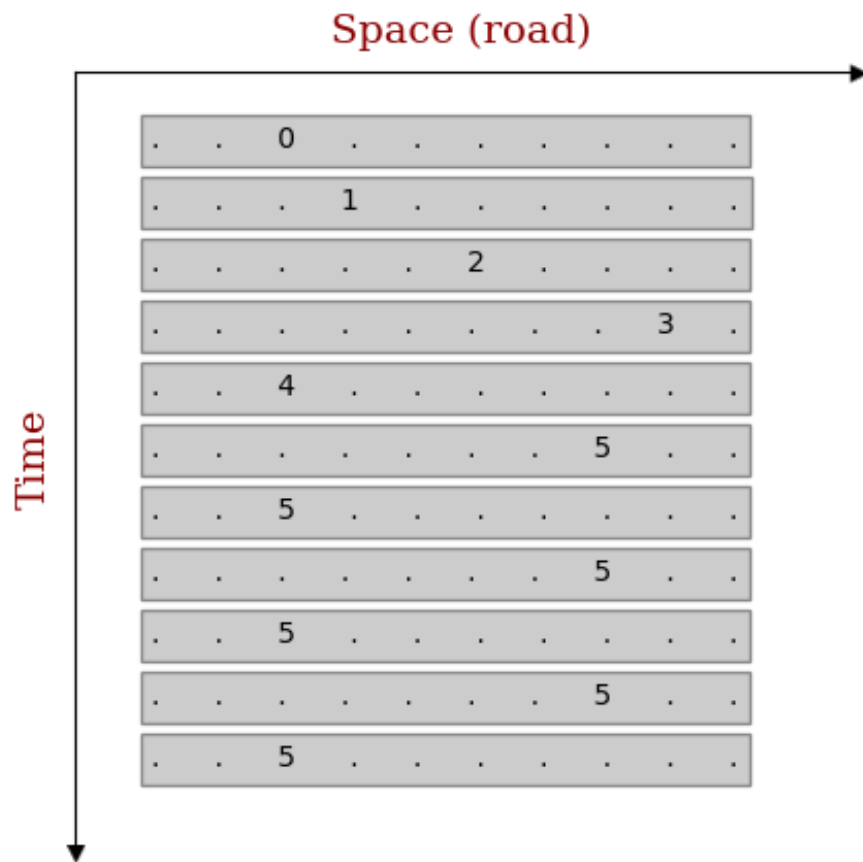


Figure 1

A simple representation of a circular road in a cellular automata simulation. The road length here is 10 and the simulation has been run for 10 seconds. The integer represents the position of a car and its magnitude represents its speed. As the car reaches the end of the line, it simply loops around in a circle. As no other rules govern its movement, it reaches the imposed speed limit on the road. All simulations will be done with a speed limit of 5 unless stated otherwise. Further rules to be added to govern traffic flow are deceleration rules due to other cars, and random probabilities of slowing down to simulate real-world traffic.

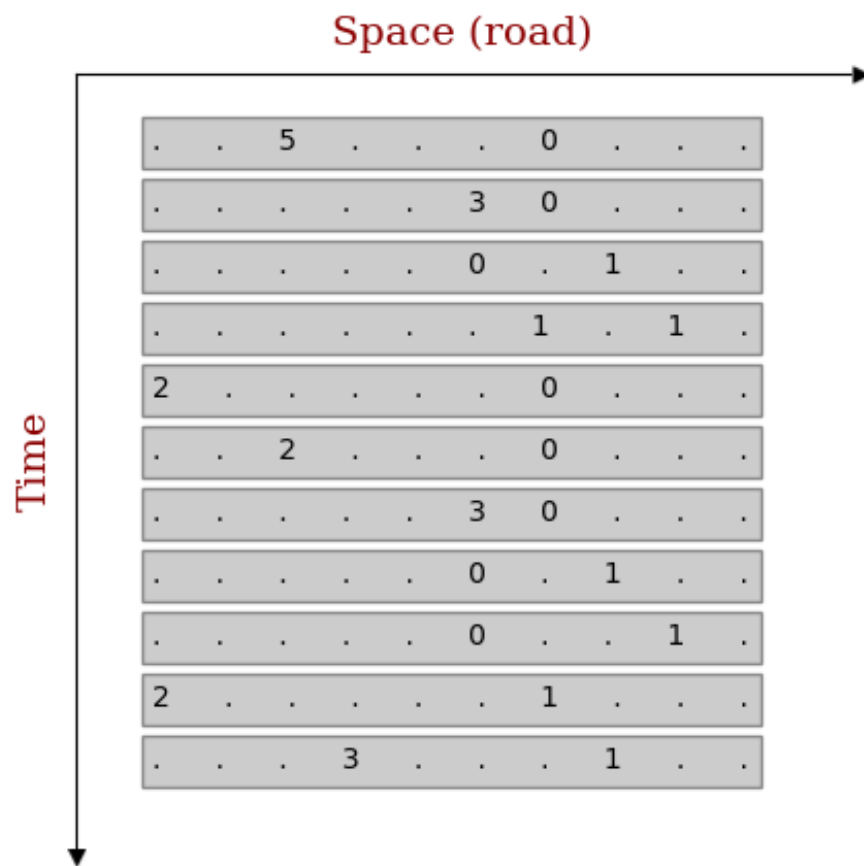


Figure 2

The test case scenario on the same road length runs for 10 seconds with more rules to cause deceleration due to other cars and random deceleration. As can be seen, a mini traffic jam forms as the second car stays stationary for the first 2 seconds.

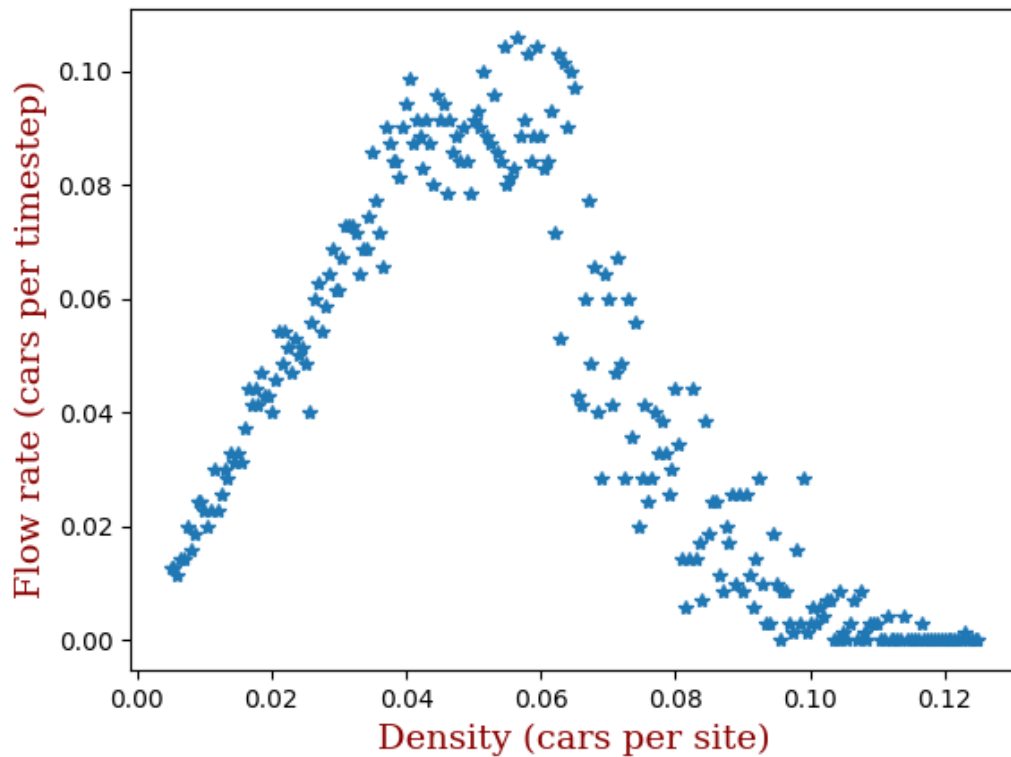
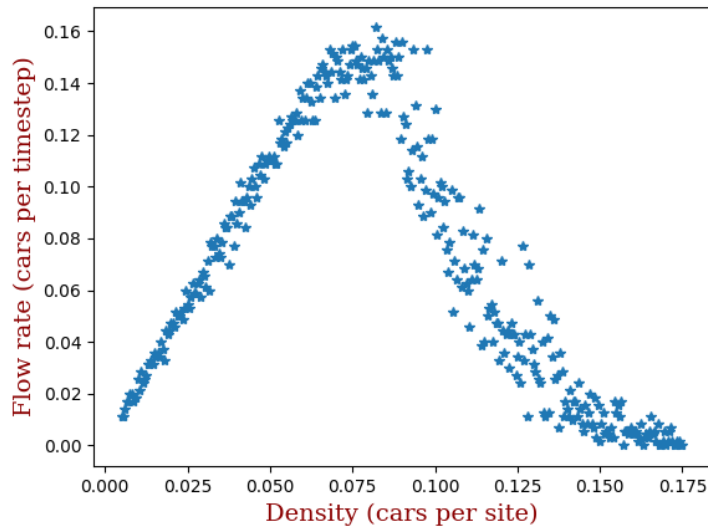
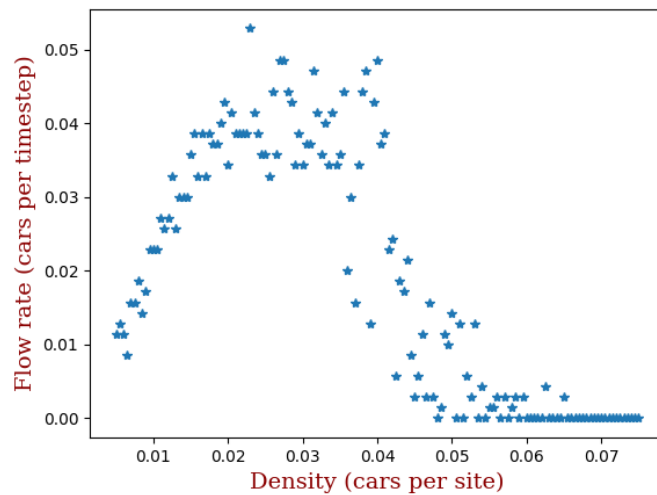


Figure 4

This is the flow rate calculation as a function of density of the cars on the road. The density of cars on the road is found by dividing the number of cars by the length of the road. The flow rate is calculated by dividing the number of cars by the number of monitored time steps. In this case, monitored time steps does not equal the entire simulation length as the cars start from stationary and need a short amount of time to accelerate and move as they would on a real road. There is a critical density at which flow rate is maximum around 0.06 cars per site, as more cars are added past this point, the flow rate decreases. This is likely due to having too many cars with random slowing probabilities causes mini traffic jam clusters. If these clusters are close together, they form larger jams and reduce the flow rate. This simulation was carried out with a road length of 2000, for 750 times steps and with 10 cars all the way up to 250 cars on the road. Running these simulations over longer road lengths and time periods as well as averaging over multiple runs would provide a more accurate representation



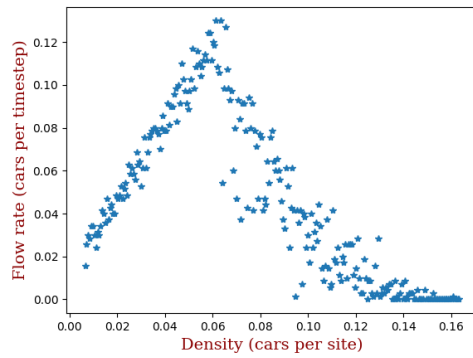
(a) Probability threshold, $p = 0.4$



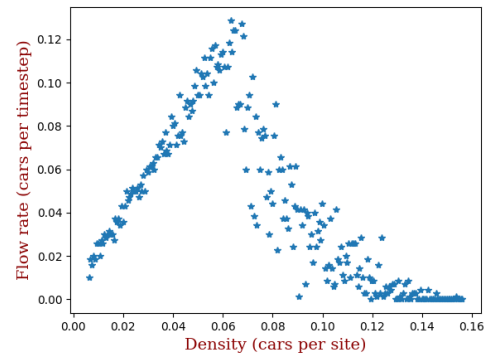
(b) Probability threshold, $p = 0.8$

Figure 5

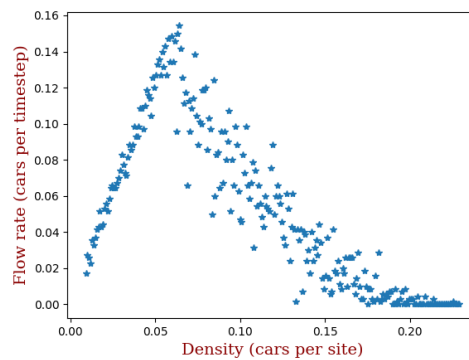
So far, simulations have been conducted with random numbers being compared to a threshold value (0.6), if the number is less than the threshold the car slows down. By changing the threshold value the critical density point can be shifted and the maximum flow rate can be changed greatly. This simulates what traffic flow would look like if drivers were more confident versus if they are more risk averse. This can be representative of traffic in different countries where driver temperament or road infrastructure can vary greatly. The number of cars must be adjusted for each variation to produce graphs that display a distribution that is easy to read.



(a) Road length = 1531



(b) Road length = 1601



(c) Road length = 1901

Figure 6

Until now, the road length being considered has been a multiple of 10 being either 10, 100 or 2000. The key observation with the usage of prime numbers for road length shows that the distribution has a sharper peak at the critical density compared to other results so far where the peak is more rounded. Flow rates and density values did not have any noticeable changes. The simulations have been run for 3 different prime numbers so there is still room for simulations to be run and averages over them to establish a correlation. As car speeds are usually not prime numbers, this could provide interesting insight into traffic flow.

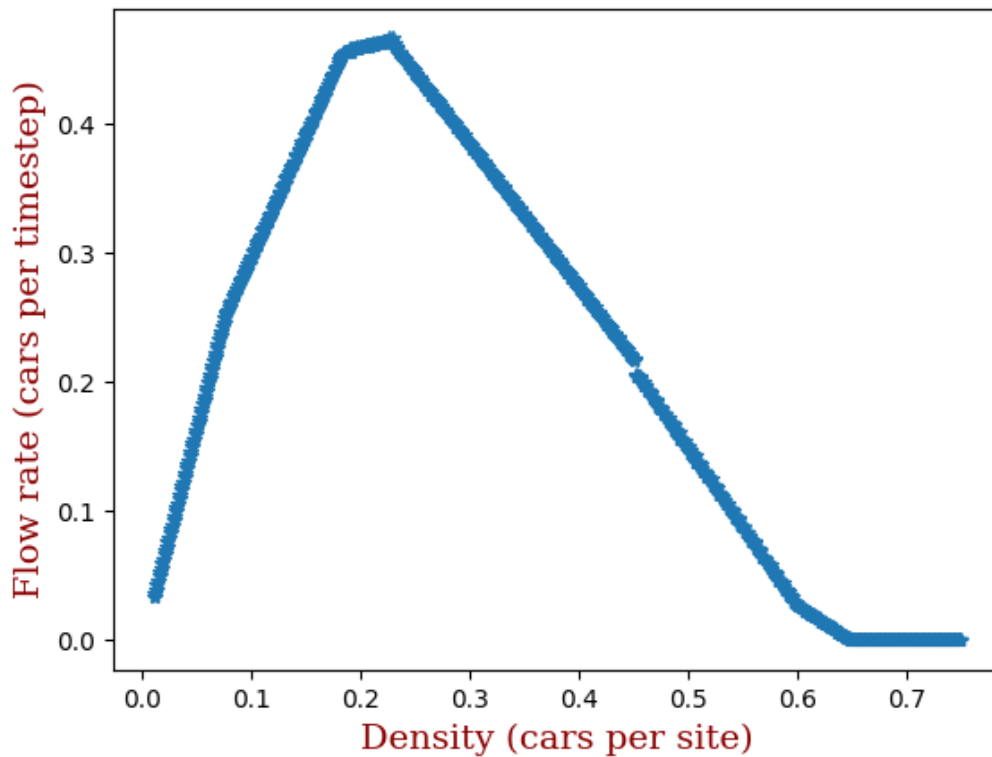


Figure 7

Running the simulation with a 'road works' region of restricted speed limit yielded interesting effects on the flow rate. It caused the points to assemble on a curve and allowed very high car density and flow rates without traffic jams building up and restricting flow. This variation of the code was run for a shorter road length of 1000 with a larger range of car numbers. As can be seen the critical road density is quite high at about 0.25, compared to the 0.6 obtained for a simulation for a simple road with longer length and less cars. This can also model a smart motorway where specific regions of the motorway are speed restricted to prevent traffic jams.

References

- [1] *Kai Nagel and Michale Schreckenberg, "A cellular automaton model for freeway traffic", J. Phys. I France 2, 1992*